23336179 李福泉 DIP第四章作业

No.4.3

<a> $\delta(t)$ 和 $\delta(t-t_0)$

$$\delta(t) * \delta(t-t_0) = \int_{-\infty}^{+\infty} \delta(\tau)\delta(t-t_0-\tau)\,d\tau$$
$$= \delta(t-t_0)$$

<b> $\delta(t-t_0) * \delta(t+t_0)$

$$= \int_{-\infty}^{+\infty} \delta(\tau-t_0)\delta(t+t_0-\tau)\,d\tau$$
$$= \delta(t+t_0-t_0)$$
$$= \delta(t)$$

No.4.9 由冲激得 $F\{e^{j2\pi u_0 t}\} = \delta(u-u_0) = \int_{-\infty}^{+\infty} e^{j2\pi(t_0-u)t}\,dt$

又由欧拉公式 $e^{j2\pi u_0 t} = \cos(2\pi u_0 t) + j\sin(2\pi u_0 t)$

$$e^{j2\pi u_0 t} = \cos(2\pi u_0 t) - j\sin(2\pi u_0 t)$$

则 $\cos(2\pi u_0 t) = \dfrac{e^{j2\pi u_0 t} + e^{-j2\pi u_0 t}}{2}$

$$\sin(2\pi u_0 t) = \frac{e^{j2\pi u_0 t} - e^{-j2\pi u_0 t}}{2j}$$

<a> $F\{\cos(2\pi u_0 t)\} = \int_{-\infty}^{+\infty} \cos(2\pi u_0 t)\, e^{-j2\pi u t}\,dt$

$$= \int_{-\infty}^{+\infty} \frac{e^{j2\pi u_0 t} + e^{-j2\pi u_0 t}}{2} e^{-j2\pi u t}\,dt$$
$$= \frac{1}{2}\int_{-\infty}^{+\infty} e^{j2\pi(u_0-u)t} + e^{-j2\pi(u_0+u)t}\,dt$$
$$= \frac{1}{2}[\delta(u-u_0) + \delta(u+u_0)]$$

<b> $F\{\sin(2\pi u_0 t)\} = \int_{-\infty}^{+\infty} \sin(2\pi u_0 t)\, e^{-j2\pi u t}\,dt$

$$= \int_{-\infty}^{+\infty} \frac{e^{j2\pi u_0 t} - e^{-j2\pi u_0 t}}{2j} e^{-j2\pi u t}\,dt$$
$$= \frac{1}{2j}\left(e^{j2\pi(u_0-u)t} - e^{-j2\pi(u_0+u)t}\right)dt$$
$$= \frac{1}{2j}[\delta(u-u_0) - \delta(u+u_0)]$$

No 4.15

(a) $F_m = \sum_{n=0}^{M-1} f_n e^{-j2\pi mn/M}$, $m = 0, 1, 2, \cdots M-1$

$f_n = \frac{1}{M} \sum_{m=0}^{M-1} F_m e^{j2\pi mn/M}$, $n = 0, 1, 2, \cdots M-1$

验证: $F_m = \sum_{n=0}^{M-1} \cdot \left[\frac{1}{M} \sum_{r=0}^{M-1} F_r e^{j2\pi rn/M}\right] e^{-j2\pi mn/M}$.

$= \frac{1}{M} \sum_{r=0}^{M-1} F_r \sum_{n=0}^{M-1} e^{j2\pi(r-m)n/M}$

$= \frac{1}{M} F_m M = F_m$.

$f_n = \frac{1}{M} \sum_{m=0}^{M-1} \sum_{r=0}^{M-1} f_r e^{j2\pi mr/M} \cdot e^{j2\pi mn/M}$

$= \frac{1}{M} \sum_{r=0}^{M-1} f_r \left[\sum_{m=0}^{M-1} e^{j2\pi mn/M} \cdot e^{-j2\pi mr/M}\right]$

$= \frac{1}{M} f_n M = f_n$ 证明完毕

(b) $F(u) = \sum_{x=0}^{M-1} f(x) e^{-j2\pi ux/M}$, $u = 0, 1, 2, \cdots M-1$

$f(x) = \frac{1}{M} \sum_{u=0}^{M-1} F(u) e^{j2\pi ux/M}$, $x = 0, 1, 2, \cdots M-1$

验证: $F(u) = \sum_{x=0}^{M-1} \left[\frac{1}{M} \sum_{r=0}^{M-1} F(r) e^{j2\pi rx/M}\right] \cdot e^{-j2\pi ux/M}$

$= \frac{1}{M} \sum_{r=0}^{M-1} F(r) \left[\sum_{x=0}^{M-1} e^{j2\pi rx/M} e^{-j2\pi ux/M}\right]$

$= \frac{1}{M} F(u) M = F(u)$

$f(x) = \frac{1}{M} \sum_{u=0}^{M-1} \left[\sum_{r=0}^{M-1} f(r) \cdot e^{j2\pi ur/M}\right] e^{j2\pi ux/M}$

$= \frac{1}{M} \sum_{r=0}^{M-1} f(r) \left[\sum_{u=0}^{M-1} e^{-j2\pi ur/M} e^{j2\pi ux/M}\right]$

$= \frac{1}{M}$ 乘 $f(x) \cdot M = f(x)$ 证毕

No 4.17

(a) $F[f(x)e^{\frac{j2\pi u_0 x}{M}}] = \sum_{x=0}^{M-1}[f(x)e^{\frac{j2\pi u_0 x}{M}}]e^{-\frac{j2\pi ux}{M}}$

$= \sum_{x=0}^{M-1} f(x)e^{-j2\pi(u-u_0)/M}$

$= F(u-u_0)$.

(b) $F^{-1}[F(u)\cdot e^{j2\pi ux_0/M}] = \frac{1}{M}\sum_{u=0}^{M-1}[F(u)e^{-j2\pi ux_0/M}]e^{j2\pi ux/M}$

$= \frac{1}{M}\sum_{u=0}^{M-1} F(u)e^{j2\pi(x-x_0)u/M}$

$= f(x-x_0)$

No 4.43 用 $f(x,y)$ 表示左图

(a) $(-1)^{x+y}\cdot f(x,y)$

(b) $F(u,v) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1}(-1)^{x+y}\cdot f(x,y)e^{-j2\pi(ux/M+vy/N)}$

(c) 取复数: 得 $F(u,v) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1}(-1)^{x+y}f(x,y)e^{-j2\pi[\frac{u(-x)}{M}+\frac{v(-y)}{N}]}$

(d) 反变换得 $f'(x,y) = \frac{1}{MN}\sum_{u=0}^{M-1}\sum_{v=0}^{N-1}F'(u,v)e^{j2\pi(ux/M+vy/N)}$

$= (-1)^{-x-y}(f(-x,-y)$

(e) 乘上 $(-1)^{x+y}$ 得 $f(-x,-y)$

所以即为中心对称变换.

No 4.47.

(a) 该空间核函数 $g(x) = \frac{1}{4}[f(x,y-1)+f(x-1,y)+f(x+1,y)+f(x,y+1)]$

$= \frac{1}{4}F(u,v)[e^{j2\pi v/N}+e^{j2\pi u/M}+e^{-j2\pi v/N}+e^{-j2\pi u/M}]$

$= H(u,v)\cdot F(u,v)$

(b) $H(u,v) = \frac{1}{4}[e^{j2\pi v/N}+e^{j2\pi u/M}+e^{-j2\pi v/N}+e^{-j2\pi u/M}]$

$= \frac{1}{2}[\cos\frac{2\pi u}{M}+\cos\frac{2\pi v}{N}\cdot)$

2. 福泉 DIP第四章作业

(b) 考虑 u取$[-\frac{M}{2}, \frac{M}{2}]$ v取$[-\frac{N}{2}, \frac{N}{2}]$

该函数在原点处取最大值，向四个方向都减小，

因此该滤波器通过直流项和低频抑制高频，

$H(u,v)$是一个低通滤波器传递函数。

(c) (1) 原因①高斯低通滤波器使中心区域低频信息平滑化

②高斯高通滤波器削弱低频成分，但中心区域对比度低，部分残余信息保留，所以明亮。

③高频增强后，书籍边缘更清晰，中心区域呈实心明亮

(d) 傅丽叶变换是线性过程，改变顺序，结果不变。